Extended weak order on Coxeter groups

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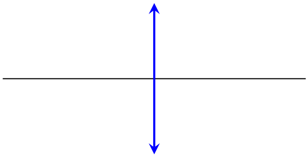
Root systems and Coxeter groups

Root systems

Let V be a real vector space with symmetric bilinear form $(-,-)$.
If $\alpha \in V$ has $(\alpha,\alpha) > 0$, then the **reflection over α** is

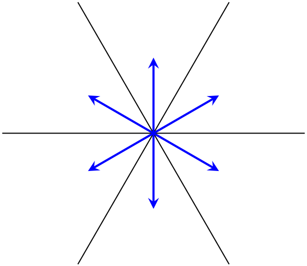
$$t_\alpha : \mathbf{v} \mapsto \mathbf{v} - 2\frac{(\alpha,\mathbf{v})}{(\alpha,\alpha)}\alpha$$



Root systems

A **finite root system** is a finite collection of vectors (called **roots**) so that

- Reflecting any root over any other root gives another root
- Any line contains at most two roots



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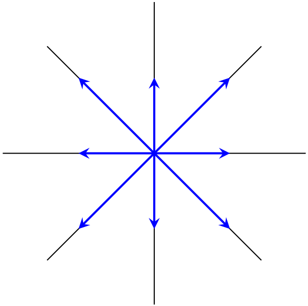

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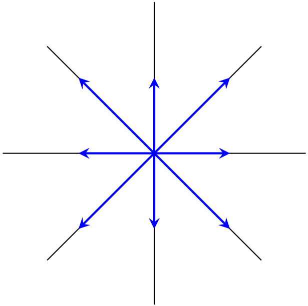

The collection of hyperplanes dual to the roots is called the **Coxeter arrangement**

General root systems

Definition

A (real) **root system** is a set $\Phi \subseteq V$ of **roots** so that:

- Each $\alpha \in \Phi$ has $(\alpha, \alpha) > 0$ and $t_\alpha \Phi = \Phi$, and

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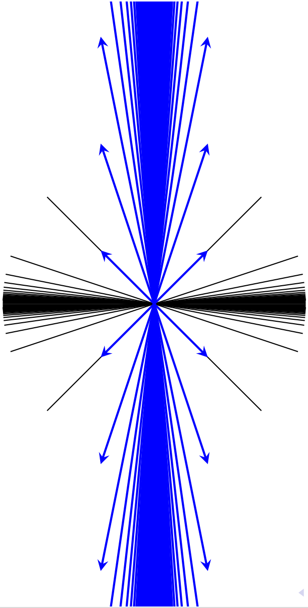
Let W be the **Weyl group** of Φ, which is the subgroup of $\mathrm{GL}(V)$ generated by $\{t_\alpha \mid \alpha \in V\}$. Then we further require:

- There exists a linearly independent set $\Pi \subseteq \Phi$ so that:
  - $\Phi = \bigcup_{w \in W} w\Pi$
  - $\Phi \subseteq \mathrm{cone}(\Pi) \cup \mathrm{cone}(-\Pi)$
- For any $\alpha \in \Phi$, $\mathbb{R}\alpha \cap \Phi = \{\pm\alpha\}$

An infinite root system: $\widetilde{A}_1$



The Weyl group of Φ is the group W generated by reflections over roots

Theorem

The Weyl groups of root systems are exactly the **Coxeter groups**: groups with a set of generators $S = \{s_i\}_{i \in I}$ so that

$$W \cong \langle S \mid (s_i s_j)^{m_{ij}} = 1,\ i, j \in I \rangle.$$

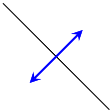
Here $m_{ij} \in \mathbb{Z}_{\geq 2} \cup \{\infty\}$, and we require $m_{ij} = m_{ji}$ and $m_{ii} = 1$.

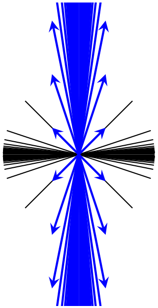
Finite Coxeter groups are classified by families A_n, $B_n = C_n$, D_n, $I_2(m)$, and exceptional groups $E_6, E_7, E_8, F_4, H_3, H_4$.

Affine root systems

Let Φ be a **crystallographic** finite root system: meaning there exists a basis of roots $\Pi = \{\alpha_1, \ldots, \alpha_n\}$ so that $\Phi \subseteq \mathbb{Z}\Pi$

Then there is an infinite root system $\widetilde{\Phi}$ called an **affine root system**, with roots of the form $\Phi + \mathbb{Z}\delta$





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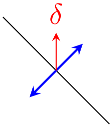
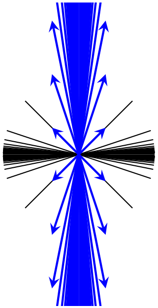


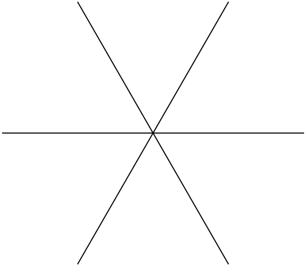


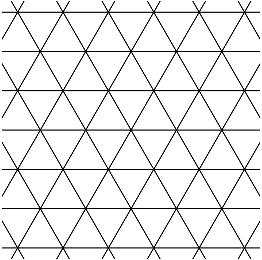
Example

The symmetric group S_n is a Coxeter group. Its root system is called the **type A_{n-1} root system**:

$$\Phi = \{\mathbf{e}_i - \mathbf{e}_j \mid i \neq j,\ i,j \in [n]\}$$

The corresponding affine root system is called the **type $\widetilde{A}_{n-1}$ root system**. Its Weyl group is the **affine symmetric group** $\widetilde{S}_n$.



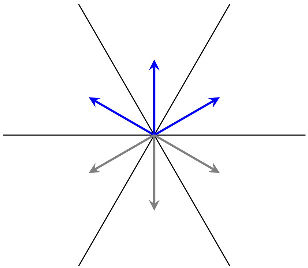


Weak order

Positive roots

Root systems can be split into **positive roots** and **negative roots**

Φ^+



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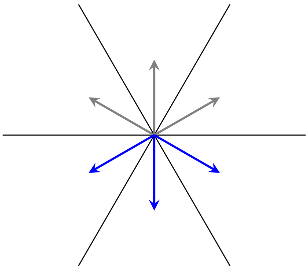

Separable sets of roots

A **separable set** of positive roots is $B \subseteq \Phi^+$ so there is a hyperplane separating B from $\Phi^+ \setminus B$.

Equivalently, there is a linear map $f : V \to \mathbb{R}$ so that $f(B) < 0$ and $f(\Phi^+ \setminus B) > 0$.

Separable sets correspond to regions of the Coxeter arrangement.

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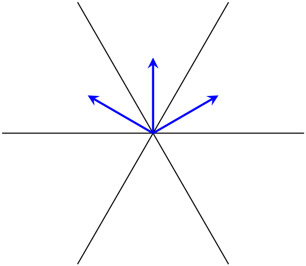

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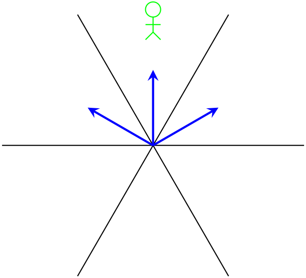

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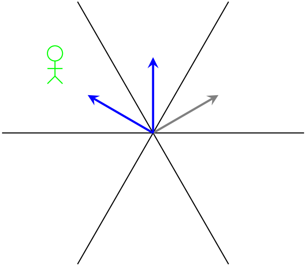

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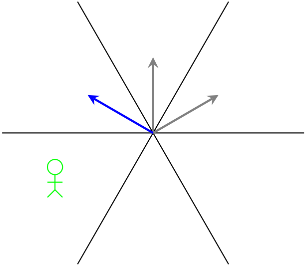

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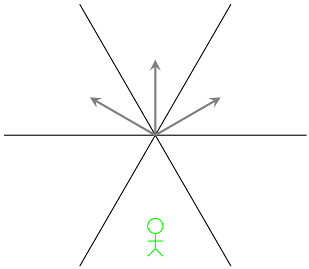

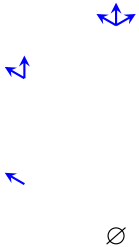

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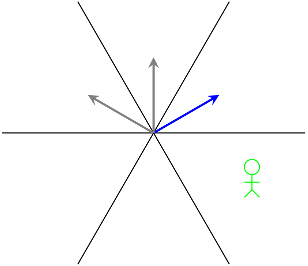

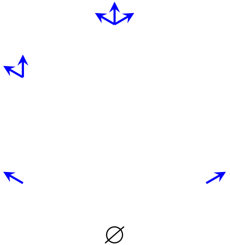

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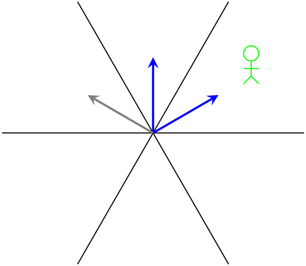

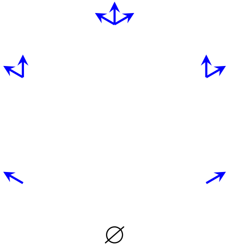

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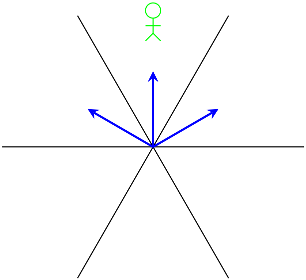
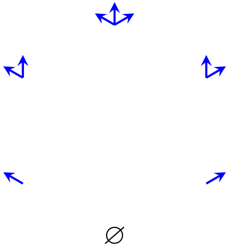




Weak order

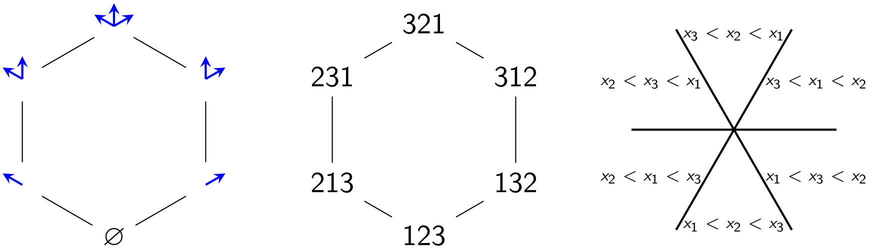
There is a separable set associated to each $w \in W$.

Definition

The **inversion set** of $w \in W$ is

$$\mathrm{Inv}(w) := \{\alpha \in \Phi^+ \mid w^{-1}\alpha \in \Phi^-\}.$$

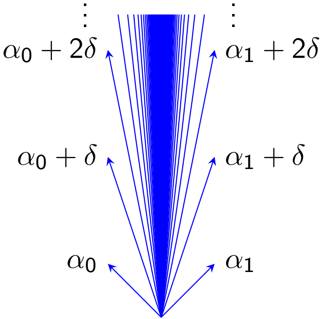
The **weak order** on W puts $w_1 \leq w_2$ if $\mathrm{Inv}(w_1) \subseteq \mathrm{Inv}(w_2)$.



Roots for $\widetilde{A}_1$

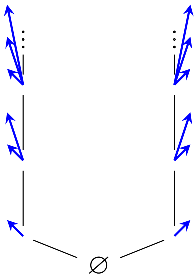
Let $\Phi = \{\pm\alpha_1\}$ be the A_1 root system. Set $\alpha_0 := \delta - \alpha_1$. The positive roots of the type $\widetilde{A}_1$ affine root system are

$$\Phi^+ = \{\alpha_0 + n\delta, \alpha_1 + n\delta \mid n \in \mathbb{Z}_{\geq 0}\}.$$



Weak order on $\widetilde{S}_2$

Inversion sets for $\widetilde{S}_2$



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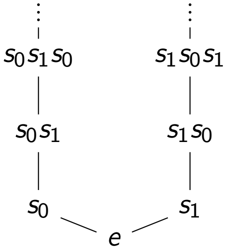
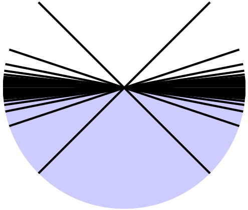


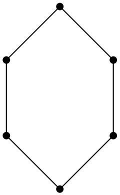
Coxeter arrangement



Definition

A partial order $\leq$ on $\mathcal{X}$ is a **complete lattice** if, for any collection $\{x_i\}_{i\in I} \subseteq \mathcal{X}$, the following two conditions hold:

1. There is a least upper bound $\bigvee_{i\in I} x_i$ of $\{x_i\}_{i\in I}$ in $\mathcal{X}$ (the **join** of $\{x_i\}_{i\in I}$)
2. There is a greatest lower bound $\bigwedge_{i\in I} x_i$ of $\{x_i\}_{i\in I}$ in $\mathcal{X}$ (the **meet** of $\{x_i\}_{i\in I}$)



The lattice property

Theorem (Björner, Björner–Edelman–Ziegler, ...)

If W is a finite Coxeter group, then the weak order on W is a complete lattice.

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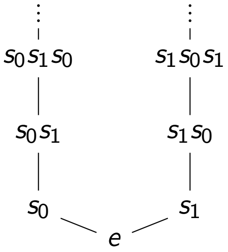

This fails when W is infinite.

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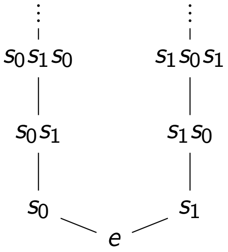

This fails when W is infinite.
Interpretation: weak order on infinite W has missing elements.
What are these missing elements?

Biclosed sets

Biclosed sets

A set $B \subseteq \Phi^+$ positive roots is **biclosed** if it satisfies the following two properties for all $\alpha, \beta, \gamma \in \Phi^+$ such that $\gamma = a\alpha + b\beta$ with $a, b \geq 0$:

(Closed) If $\alpha \in B$ and $\beta \in B$, then $\gamma \in B$.

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Example: Biclosed sets for S_3

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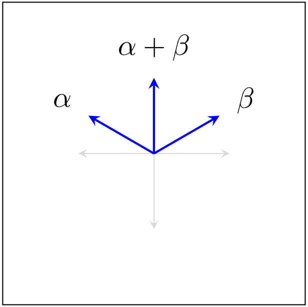

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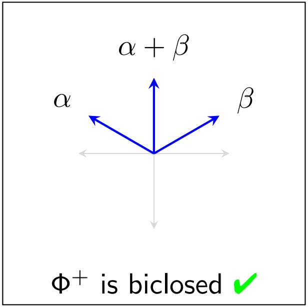

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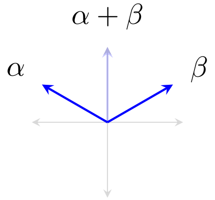

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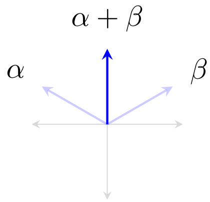

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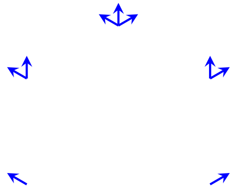
Biclosed sets in Φ^+:



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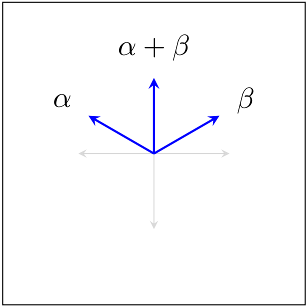

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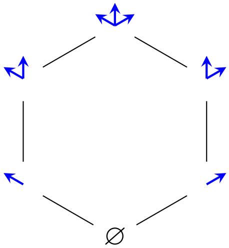
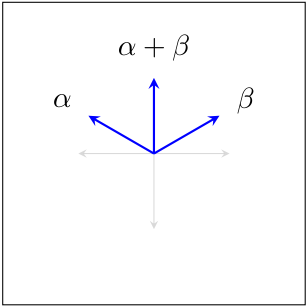




Biclosed sets are locally separable

In dimension 2, the biclosed sets coincide with the separable sets.

Fact

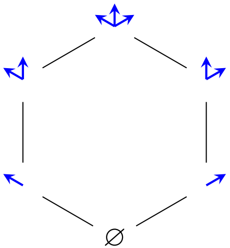
$B \subseteq \Phi^+$ is biclosed iff for any 2-dimensional space V', the restriction $B \cap V'$ is a separable set in $\Phi^+ \cap V'$.

Definition

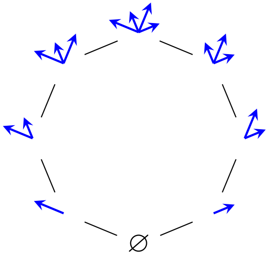
If Φ^+ is the set of positive roots for a Coxeter group W, then the poset $\mathrm{Bic}(\Phi^+)$ consisting of biclosed sets, ordered by containment, is called the **extended weak order** of W.

Extended weak order

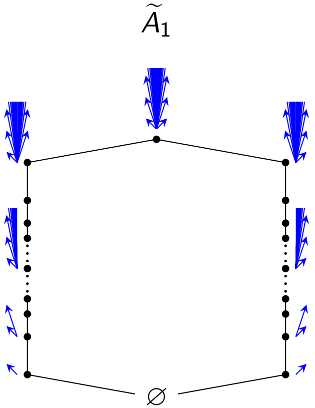
A_2



B_2



Extended weak order

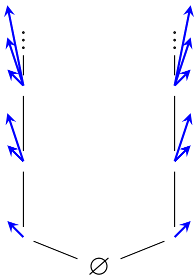


Why do biclosed sets extend weak order?

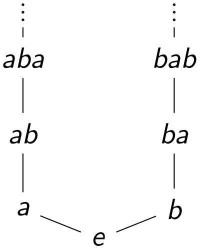
Theorem (Dyer, Cellini–Papi, Ito, ...)

If Φ is a root system, then the inversion sets $\mathrm{Inv}(w)$ for $w \in W$ are exactly the *finite* biclosed sets in Φ^+. These correspond to regions of the Coxeter arrangement in the **Tits cone**.

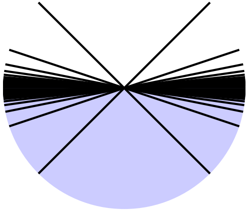
Finite biclosed sets in Φ^+



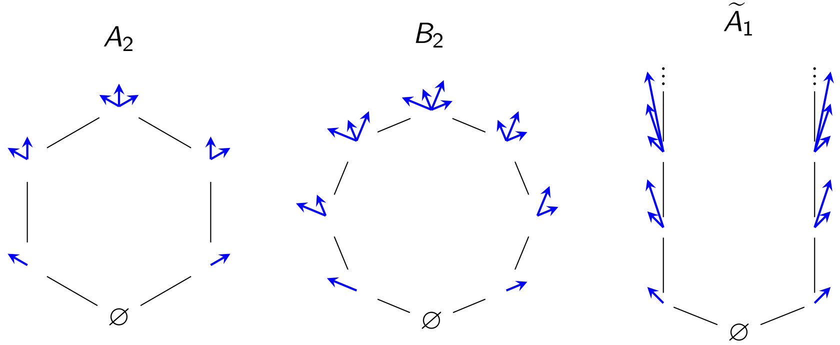
Weak order on $\widetilde{S}_2$



Coxeter arrangement for Φ



By the theorem, the collection of finite biclosed sets (an order ideal in extended weak order) is isomorphic to weak order on W



History

- Finite separable sets well-studied (Coxeter complex)
- Other separable sets also studied:
  - Borel subalgebras of Kac–Moody algebras
  - stability conditions for quiver representations
- Biclosed sets introduced by Matthew Dyer and Tom Edgar:
  - reflection orders
  - twisted Bruhat orders
  - twisted Kazhdan–Lusztig polynomials

Dyer's main conjectures

Let W be any Coxeter group.

Dyer's Conjecture A

If $B_1 \lessdot B_2$ is a cover relation in extended weak order, then

$$|B_2 \setminus B_1| = 1.$$

Dyer's Conjecture B

The extended weak order of W is a complete lattice.

Why not use separable sets?

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Fact

If $B_1 \lessdot B_2$ is a cover relation in extended weak order, and both are separable sets, then $|B_2 \setminus B_1| = 1$.

Question: Why use biclosed sets instead of separable sets then?

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If $B_1 \lessdot B_2$ is a cover relation in extended weak order, and both are separable sets, then $|B_2 \setminus B_1| = 1$.

Question: Why use biclosed sets instead of separable sets then?
One answer: In any affine Coxeter group of rank ≥ 4, the separable sets fail Dyer's Conjecture B: they still have missing elements!

Progress

Dyer's Conjecture A

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Theorem (Dyer, B–Speyer 2023)

Dyer's Conjecture A is true for affine Coxeter groups.

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Theorem (B–Defant–Hersh–McCammond–McConville–Speyer 2025, 2026+)

Dyer's Conjecture A is true for rank 3 Coxeter groups.

Progress

Dyer's Conjecture B

For any root system Φ, extended weak order is a complete lattice.

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Theorem (Björner 1984)

The finite biclosed sets (i.e. the weak order) for any Coxeter group form a meet-semilattice.

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Dyer's Conjecture B is true for the rank 3 affine Coxeter groups $\widetilde{A}_2$, $\widetilde{C}_2$, and $\widetilde{G}_2$.

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Clean arrangements

Definition

Let $\mathcal{A}$ be a hyperplane arrangement, and let X be the set of normal vectors to the hyperplanes pointing towards some region. (For instance, $\mathcal{A}$ is a Coxeter arrangement and X is the positive roots.)

We say that $\mathcal{A}$ is a **clean arrangement** if every biclosed set in X is a separable set.

Theorem (B–Speyer 2023)

Every "ideal subarrangement" of a finite or rank 3 affine Coxeter arrangement is a clean arrangement.

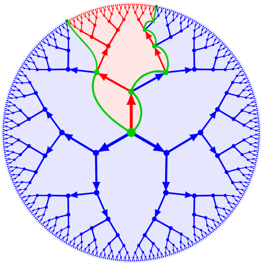
Theorem (Yoshinaga 2024)

Every finite $K(\pi, 1)$ arrangement is a clean arrangement.

Rank 3 cleanliness: snakes on a hyperbolic plane

Theorem (B–Defant–Hersh–McCammond–McConville–Speyer 2025, 2026+)

Every rank 3 Coxeter arrangement is a clean arrangement.



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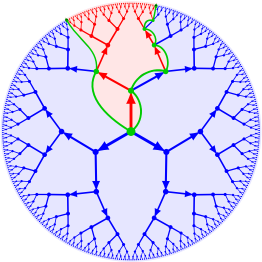

Rank 4 affine Coxeter arrangements are not clean!

Biclosed sets for S_∞ and $\widetilde{A}_{n-1}$

An example: S_∞

Let S_∞ be the group of permutations of $\mathbb{Z}$ fixing all but finitely many integers

S_∞ is a Coxeter group. What does extended weak order look like for it?

Observation

The weak order on permutations of $\mathbb{Z}$ does not form a complete lattice.
Example issue: Joining

$$\pi_0 = \ldots, -3, -2, -1, \mathbf{0}, 1, 2, 3\ldots$$
$$\pi_1 = \ldots, -3, -2, \mathbf{0}, -1, 1, 2, 3\ldots$$
$$\pi_2 = \ldots, -3, \mathbf{0}, -2, -1, 1, 2, 3\ldots$$
$$\pi_3 = \ldots, \mathbf{0}, -3, -2, -1, 1, 2, 3\ldots$$
$$\vdots$$

gives us an object in which 0 is inverted with all negative integers.
We might expect this to look like

$$0, \ldots, -3, -2, -1, 1, 2, 3, \ldots$$

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gives us an object in which 0 is inverted with all negative integers. We might expect this to look like

$$0, ..., -3, -2, -1, 1, 2, 3, ...$$

This isn't a permutation! But it is an ordering of the integers.

Idea

Definition

Let $\lhd$ be a total order of $\mathbb{Z}$. An **inversion** of $\lhd$ (with respect to $<$) is a pair (i, j) such that

$$i < j$$
$$i \rhd j$$

If we associate to a permutation of $\mathbb{Z}$ the total ordering

$$\ldots \lhd \pi(0) \lhd \pi(1) \lhd \pi(2) \lhd \ldots$$

this coincides with inversions for the permutation.

Weak order for total orders?

Let $\mathrm{Tot}(\mathbb{Z})$ be the set of all total orders on $\mathbb{Z}$.

Definition

The **inversion set** for a total order $\lhd$ (with respect to $<$) is

$$\mathrm{Inv}(\lhd) = \{(i, j) \mid i < j,\ i \rhd j\}.$$

The **weak order** on $\mathrm{Tot}(\mathbb{Z})$ is given by containment of inversion sets.

Theorem

The weak order on $\mathrm{Tot}(\mathbb{Z})$ is a complete lattice.

How do total orders relate to biclosed sets?

Theorem (B–Speyer 22)

The map

$$\begin{aligned}\mathrm{Tot}(\mathbb{Z}) &\to 2^{\Phi^+} \\ \lhd &\mapsto \{\mathbf{e}_i - \mathbf{e}_j \mid (i,j) \in \mathrm{Inv}(\lhd)\}\end{aligned}$$

is an order isomorphism from $\mathrm{Tot}(\mathbb{Z})$ to $\mathrm{Bic}(S_\infty)$.

Affine permutations

Definition

The **affine symmetric group** $\widetilde{S}_n$ is the group of bijections $\widetilde{\pi} : \mathbb{Z} \to \mathbb{Z}$ satisfying:

(a) $\widetilde{\pi}(i+n) = \widetilde{\pi}(i) + n$ for all $i \in \mathbb{Z}$, and

(b) $\displaystyle\sum_{i=1}^{n} \widetilde{\pi}(i) = \sum_{i=1}^{n} i$.

$\widetilde{S}_n$ is the Coxeter group of type $\widetilde{A}_{n-1}$.

The type $\widetilde{A}_{n-1}$ root system

Let $\widetilde{V}$ be the vector space spanned by vectors $\widetilde{\alpha}_{i,j}$ with $i, j \in \mathbb{Z}$, subject to the relations

$$\widetilde{\alpha}_{i,j} + \widetilde{\alpha}_{j,k} = \widetilde{\alpha}_{i,k}$$

and

$$\widetilde{\alpha}_{i,j} = \widetilde{\alpha}_{i+n,j+n}$$

for all $i, j, k \in \mathbb{Z}$.

The root system of $\widetilde{S}_n$ is

$$\widetilde{\Phi} = \{\widetilde{\alpha}_{i,j} \mid i \not\equiv j \bmod n\}.$$

Given a total order $\lhd$ on $\mathbb{Z}$, define

$$I(\lhd) = \{\widetilde{\alpha}_{i,j} \mid (i,j) \in \mathrm{Inv}(\lhd),\ i \not\equiv j \bmod n\}.$$

We say $\lhd$ is **translation invariant** if, for all $i,j \in \mathbb{Z}$,

$$i \lhd j \quad \text{iff} \quad i+n \lhd j+n.$$

Theorem (B–Speyer 2022)

A set $B \subseteq \widetilde{\Phi}^+$ is biclosed if and only if there is some translation invariant total order $\lhd$ such that $B = I(\lhd)$.

This leads to a very explicit parametrization of $\mathrm{Bic}(\widetilde{A}_{n-1})$, using **window notation**.

Applications of biclosed sets

Application 1: Coxeter matroid polytopes

Let W be a Coxeter group with root system $\Phi \subseteq V$

Definition

Let $\rho \in V$ be a vector not fixed by reflection over any root. A finite subset $X \subseteq W$ is a **Coxeter matroid** if every edge of the polytope

$$\mathrm{Pol}_X = \mathsf{conv}(\{x\rho \mid x \in X\})$$

is parallel to a root.

Let G_X be the 1-skeleton of Pol_X, with edges directed in the negative root direction.

Conjecture

For any Coxeter matroid X and biclosed set $B \subseteq \Phi^+$, reverse the edges of G_X which are parallel to roots in B. Then the resulting directed graph is acyclic and has a unique source and sink.

Application 2: Bruhat decompositions

Conjecture

For any Coxeter matroid X and biclosed set $B \subseteq \Phi^+$, reverse the edges of G_X which are parallel to roots in B. Then the resulting directed graph is acyclic and has a unique source and sink.

Corollary

For a Kac–Moody group G with Borel subgroup B, and any biclosed set $A \subseteq \Phi^+$, there is a "twisted Bruhat decomposition"

$$G = \bigsqcup_{w \in W} U_w w B$$

where U_w is a product of root subgroups depending on A and w.

Application 2: Bruhat decompositions

Conjecture

For any Kac–Moody algebra $\mathfrak{g}$ with roots Φ, and any biclosed set $A \subseteq \Phi^+$, there is a subalgebra $\mathfrak{a} \subseteq \mathfrak{g}$ such that the real root space $\mathfrak{g}_\alpha \subseteq \mathfrak{a}$ if and only if $\alpha \in A$ or $-\alpha \in \Phi^+ \setminus A$.

This would mean biclosed sets are exactly the positive roots appearing in some generalized Borel subalgebra.

Corollary

The twisted Bruhat decomposition

$$G = \bigsqcup_{w \in W} U_w w B$$

is the decomposition into orbits of a generalized Borel subgroup of G acting on G/B.

Application 3: Cluster algebras

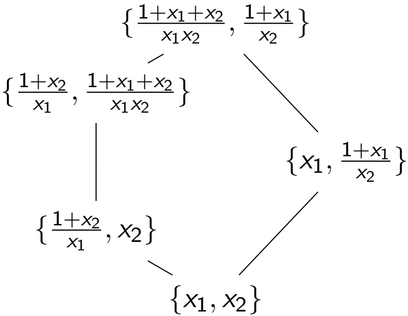
Associated to each (loopless 2-acyclic) quiver Q is a **cluster algebra** A_Q.

$$Q = \bullet \longrightarrow \bullet$$

Each node has a variable x_i attached

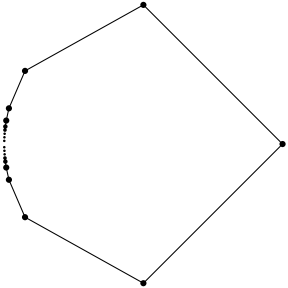
Clusters are sets of Laurent polynomials in $x_1, \ldots, x_n$ built from $\{x_1, \ldots, x_n\}$ using **mutation**

The **exchange graph** has vertices the clusters and edges the mutations



Now consider the "type $\widetilde{A}_1$" quiver

$$Q = \bullet \rightrightarrows \bullet$$



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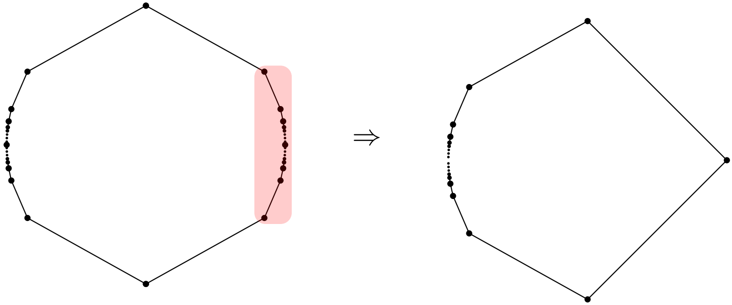

Application 3: Cluster algebras

Theorem (B–Defant 2024, B 2025)

If Q is any orientation of an affine Dynkin diagram with affine Coxeter group W, then there is a quotient of the extended weak order of W which realizes the ordered exchange graph of A_Q.

These are called **affine Cambrian quotients**

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These are called **affine Cambrian quotients**

Conjecture

If Q is any acyclic quiver, and one interprets Q as a Coxeter–Dynkin diagram with Weyl group W, then there is a quotient of the extended weak order of W which realizes the ordered exchange graph of A_Q.

Application 4: Torsion classes

A **torsion class** for an algebra Π is a collection of finite-dimensional Π-modules which is closed under quotients and extensions.
Torsion classes form a lattice under containment.

Theorem (B 2025)

If W is a (simply-laced) affine Coxeter group, then the extended weak order of W is a quotient of the lattice of torsion classes of a **preprojective algebra**.

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If W is a (simply-laced) affine Coxeter group, then the extended weak order of W is a quotient of the lattice of torsion classes of a **preprojective algebra**.

Conjecture

If W is any (crystallographic) Coxeter group, then the extended weak order of W is a quotient of the lattice of torsion classes of a preprojective algebra Π_W.

Thank you!

Definition

The **affine symmetric group** $\widetilde{S}_n$ is the group of bijections $\widetilde{\pi} : \mathbb{Z} \to \mathbb{Z}$ satisfying:

(a) $\widetilde{\pi}(i+n) = \widetilde{\pi}(i) + n$ for all $i \in \mathbb{Z}$, and

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Condition (b) lets us recover $\widetilde{\pi}$ from one-line notation: e.g. the identity is

$$\ldots, -3, -2, -1, 0, 1, 2, 3, 4, 5, \ldots$$

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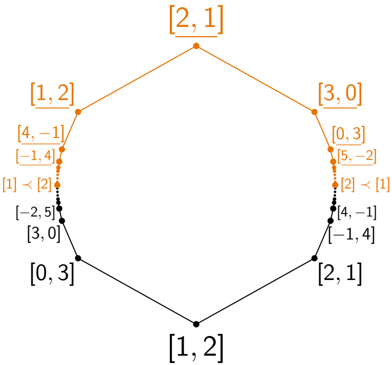
We can also encode an affine permutation via **window notation**: e.g. in $\widetilde{S}_3$

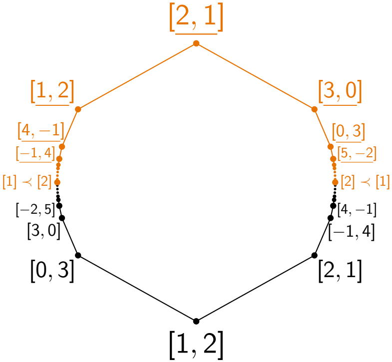
$$[0, 1, 2] = [1, 2, 3] = \mathrm{id},$$

$$[1, 0, 2] = \ldots, -2, -3, -1, 1, 0, 2, 4, 3, 5, \ldots$$

Extended weak order



Extended weak order



The black vertices are labeled with elements of $\widetilde{S}_n$.
What are the other labels?

Definition (B.–Speyer 2022)

A **translation invariant total order** (**TITO**) is a total order ($\prec$) on $\mathbb{Z}$ satisfying the following property:

- For all $i, j \in \mathbb{Z}$, we have $i \prec j$ if and only if $i + n \prec j + n$.

A TITO is **real** if it further satisfies:

- For all $i \in \mathbb{Z}$, if $i + n \prec i$ then there exists a k with $i + n \prec k \prec i$.

TITOs can be encoded with window notation: e.g. in $\widetilde{\mathcal{B}}_3$ the notation $[1] \prec [3, 2]$ encodes

$$\cdots \prec -2 \prec 1 \prec 4 \prec 7 \prec \cdots \prec \cdots \prec 0 \prec -1 \prec 3 \prec 2 \prec 6 \prec 5 \prec \cdots$$

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We write $[1] \prec [\underline{3, 2}]$ to instead encode

$$\cdots \prec -2 \prec 1 \prec 4 \prec 7 \prec \cdots \prec \cdots \prec 6 \prec 5 \prec 3 \prec 2 \prec 0 \prec -1 \prec \cdots$$

Theorem (B.–Speyer 2022)

The map $\lhd \mapsto I(\lhd)$ is a bijection from real TITOs to $\mathrm{Bic}(\widetilde{A}_{n-1})$.

An example

Consider the sequence in $\widetilde{S}_2$:

$$\cdots \prec -5 \prec -4 \prec -3 \prec -2 \prec -1 \prec \textcolor{red}{0} \prec 1 \prec 2 \prec 3 \prec 4 \prec 5 \prec \cdots$$
$$\cdots \prec -5 \prec -2 \prec -3 \prec \textcolor{red}{0} \prec -1 \prec 2 \prec 1 \prec 4 \prec 3 \prec 6 \prec 5 \prec \cdots$$
$$\cdots \prec -5 \prec \textcolor{red}{0} \prec -3 \prec 2 \prec -1 \prec 4 \prec 1 \prec 6 \prec 3 \prec 8 \prec 5 \prec \cdots$$
$$\vdots$$

The join in extended weak order is

$$\cdots \prec -4 \prec -2 \prec \textcolor{red}{0} \prec 2 \prec 4 \prec \cdots \prec -3 \prec -1 \prec 1 \prec 3 \prec 5 \prec \cdots$$

An example